FLUID–STRUCTURE INTERACTION OF THREE-DIMENSIONAL PRINTED FLEXIBLE UNDERWATER VEHICLE FINS: DUAL-FIDELITY MODELING AND DEFORMED-CAMBER VERIFICATION

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Master of Technology Project Report

DEPARTMENT OF OCEAN ENGINEERING AND NAVAL ARCHITECTURE
INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

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by

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DEPARTMENT OF OCEAN ENGINEERING AND NAVAL ARCHITECTURE
INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR
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CERTIFICATE

This is to certify that the Dissertation Report entitled, "**Fluid–Structure Interaction of Three-Dimensional Printed Flexible Underwater Vehicle Fins: Dual-Fidelity Modeling and Deformed-Camber Verification**" submitted by Mr. **Sahil Patel** (Roll No. 22NA30028) to Indian Institute of Technology Kharagpur, India, is a record of bonafide Project work carried out by him under my supervision and guidance and is worthy of consideration for the award of the degree of Master of Technology in Ocean Engineering and Naval Architecture of the Institute.

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List of Symbols

Symbol	Explanation	SI Units
L, b	Hydrofoil chord length and span	m
$\tau = t/c$	Maximum thickness-to-chord ratio of NACA 00xx section	–
h_{mean}	Chordwise area-averaged thickness ($0.68508\,\tau L$)	m
$f_{\text{clamp}}, L_{\text{bend}}$	Root clamp fraction (0.12) and free bending length	–, m
U_∞	Free-stream water velocity	m/s
$\alpha_0, \alpha_{\text{eff}}$	Geometric angle of attack and effective washout angle	deg
ρ_f, ν_f	Fluid density and kinematic viscosity	kg/m^3, m^2/s
ρ_s, ν_s	Solid polymer density and Poisson ratio	kg/m^3, –
E_0, E_{eff}	Base solid modulus and homogenized sandwich modulus	Pa
$t_{\text{shell}}, \phi_{\text{infill}}$	Perimeter shell thickness and internal infill fraction	m, –
k_{orient}	Build orientation stiffness factor (`edge`, `flat`, `upright`)	–
Re	Chord Reynolds number ($U_\infty L/\nu_f$)	–
μ	Structure-to-fluid mass ratio ($\rho_s h_{\text{mean}}/(\rho_f L)$)	–
K_B, C_Y	Dimensionless bending rigidity and Cauchy number ($1/K_B$)	–
U_r	Reduced velocity ($U_\infty/(f_{1,\text{water}} L_{\text{bend}})$)	–
$f_{1,\text{vac}}, f_{1,\text{water}}$	First bending natural frequency in vacuum and in water	Hz
M_a, Γ_{3D}	Potential-flow strip added mass and finite-span factor	kg/m, –
C_L, C_D, C_M	Hydrodynamic lift, drag, and quarter-chord moment coefficients	–
u_{tip}	Trailing-edge transverse tip deflection	mm
σ_{vM}	Root surface von Mises stress	MPa
$\kappa(x)$	Nonlinear Euler–Bernoulli elastica curvature	m^{-1}
$y^+, \Delta y_1$	Non-dimensional wall distance and first prism-layer height	–, m
$\mathbf{V}_k, \mathbf{W}_k$	Secant residual and displacement difference matrices in IQN-ILS	–

Abstract

Autonomous underwater vehicles increasingly use three-dimensional printed polymer control fins, yet these compliant surfaces deform under hydrodynamic load and cannot be analyzed as rigid hydrofoils. When placed at an angle of attack, lift bends the flexible trailing edge toward the incoming flow. This passive washout reduces the effective angle of attack, sheds excess lift, lowers drag, and limits root bending stress. Modeling this interaction presents a severe numerical stability problem: because polymer density ($\rho_s \approx 1020$–$1270\ \mathrm{kg/m^3}$) is nearly equal to water density ($\rho_f \approx 998$–$1025\ \mathrm{kg/m^3}$), the structure-to-fluid mass ratio for a thin NACA 0008 section is $\mu = \rho_s h_{\mathrm{mean}}/(\rho_f L) \approx 0.066$. The entrained water contributes roughly twelve times the inertia of the fin itself, causing explicit fluid–structure coupling schemes to diverge unconditionally.

This report develops a dual-fidelity fluid–structure interaction framework in which a full-order continuum solver and a fast coupled reduced-order model are developed together. The full-order path uses Gmsh boundary-layer meshing, OpenFOAM k–ω SST unsteady Reynolds-averaged Navier–Stokes CFD, and three-dimensional nonlinear CalculiX finite-element analysis (`C3D8I`, `NLGEOM`) on prescribed rigid and deformed-camber geometries. The reduced-order path is a 0.51 s coupled C^1 Cubic Hermite model stabilized by multi-secant Interface Quasi-Newton Inverse Least-Squares (IQN-ILS), built to carry the same physics into rapid design studies. For a TPU 95A NACA 0008 benchmark at $U_\infty = 1.2$ m/s and $\alpha_0 = 8^\circ$, the full-order solver shows that bending reduces C_L from 0.9301 to 0.6414 at $t = 0.5$ s (0.6624 after steady-time extrapolation) and reduces C_D from 0.1140 to 0.0348. Intersecting the OpenFOAM fluid secant with the CalculiX compliance line yields a quasi-steady equilibrium estimate of $u^*_{\mathrm{tip}} = 2.398$ mm, $C^*_L = 0.6775$, and $\sigma^*_{\mathrm{vM}} = 0.146$ MPa. Against this reference the reduced-order model differs by 5.4% in tip deflection, 1.5% in lift, and 1.4% in root stress while running 1,150× faster. A 15-case full-order campaign across six polymers then defines the operating envelope, showing where the reduced-order model reproduces continuum behaviour and where high-speed soft elastomers demand the full solver.

Keywords: Fluid–structure interaction; Flexible underwater fins; Added-mass instability; Passive hydroelastic washout; Three-dimensional printed polymers; IQN-ILS; OpenFOAM; CalculiX.

Chapter 1

Introduction and Literature Review

1.1 Background and Physical Motivation

Unlike stiff metallic marine rudders or propellers, three-dimensional (3D) printed polymer underwater fins deform passively under water pressure. When a flexible fin operates at a geometric angle of attack α_0, hydrodynamic pressure bends the compliant trailing edge upward toward the incoming streamline. This passive deformation reduces the local angle of attack and introduces negative camber—a mechanism known as **passive hydroelastic washout**. In marine propulsion and control surfaces, controlled washout sheds excess lift during gusts, lowers pressure drag, and protects the root mounting spar from high bending stress [1, 2].

Predicting the coupled hydroelastic response of 3D-printed polymer fins requires resolving three engineering difficulties:

1. **Added-Mass Instability at Low Mass Ratio ($\mu \ll 1$)**: Standard 3D-printing polymers ($\rho_s \approx 1020$–1270 kg/m^3) have nearly the same density as water ($\rho_f \approx 998$–1025 kg/m^3). For a thin NACA 0008 hydrofoil ($t/c = 0.08$), the structure-to-fluid mass ratio is $\mu = \rho_s h_{\text{mean}}/(\rho_f L) \approx 0.066$. Because the cylinder of accelerated water weighs roughly twelve times more than the fin itself, loosely coupled or explicit fluid–structure interaction (FSI) schemes diverge within one to two time steps regardless of time-step size [3].

2. **Fused Filament Fabrication Anisotropy and Moisture Conditioning**: Printed fins behave as sandwich beams with solid outer perimeter shells and internal lattice infill whose flexural stiffness depends on build orientation (`edge`, `flat`, `upright`). In addition, polyamide (PA12 Nylon) absorbs water when submerged and loses more than 50% of its dry elastic modulus, whereas thermoplastic polyurethanes (TPU 95A and TPU 85A) undergo large deflections requiring nonlinear elastica and hyperelastic (`NLGEOM`) mechanics [4].

3. **Balancing Interactive Design Speed with Continuum Verification**: Screening dozens of polymer grades, infill fractions, print orientations, and flow velocities requires a sub-second solver, whereas engineering sign-off requires continuum Navier–Stokes computational fluid dynamics (CFD) and 3D finite-element analysis (FEA).

1.2 Literature Review

The analytical foundation for submerged flat-plate and cantilever hydroelasticity was established by Chu [5] and experimentally verified by Lindholm et al. [6], who showed that entrained fluid added mass reduces the natural bending frequencies of submerged cantilevers according to the ratio $\rho_f L/(\rho_s h)$. In computational FSI, Causin et al. [3] proved mathematically that partitioned Dirichlet–Neumann schemes diverge unconditionally whenever the fluid added-mass operator exceeds structural inertia. To stabilize partitioned coupling without assembling a monolithic fluid–structure matrix, Degroote et al. [7] introduced the Interface Quasi-Newton with Inverse Least-Squares (IQN-ILS) method, which is now implemented in the open-source coupling library preCICE [8] and verified against standard benchmarks such as Turek and Hron [9]. In marine applications, Connell and Yue [1] and Young [2] demonstrated that passive bending and twist in compliant foils and composite blades alter vortex shedding and delay cavitation and stall.

1.3 Explicit Statement of Student Contributions

The work is organised around two solvers that address the same physics at different cost. The specific contributions of the candidate are stated explicitly below:

1. Formulation and implementation of a **dimensionless governance gate** that computes Re, μ, K_B, C_Y, and U_r prior to solver execution, automatically enforcing $y^+ \approx 1$ boundary-layer meshing ($\Delta y_1 = 16.6\ \mu$m), turbulence closure selection (`laminar` versus Menter's k–ω SST [10]), and implicit `IQN-ILS` coupling.

2. Development and execution of a **full-order continuum solver** coupling parametric Gmsh 4.8.4 meshing, 8-rank MPI OpenFOAM (`pimpleFoam`) k–ω SST CFD, and 3D nonlinear CalculiX 2.17 (`C3D8I`, `*DLOAD`, `NLGEOM`) FEA, used to compute fluid loading on rigid and deformed-camber shapes and the resulting 3D deformation and root stress on a 60-vCPU cloud node.

3. Development of a 0.51 **s coupled C^1 Cubic Hermite reduced-order model** carrying the same physical mechanisms, incorporating sandwich shell–infill homogenization, nonlinear elastica curvature, Theodorsen unsteady plunge damping, C^1 rational streamline washout saturation, and rank-revealing `IQN-ILS` acceleration.

4. Construction of a **quasi-steady secant equilibrium estimate** from the computed full-order states, followed by a 15-case multi-polymer campaign that establishes a quantitative operating envelope for the reduced-order model across Cauchy number C_Y.

The remainder of this report separates **prediction** (via the coupled ROM), **continuum**

computation (via fixed-geometry OpenFOAM and CalculiX runs), and **equilibrium estimation** (via secant intersection). The distinction is maintained so that no stage is described indiscriminately as a coupled solver.

Chapter 2

Theoretical Study and Dual-Fidelity Methodology

2.1 Two Solvers for the Same Physics

Figure 2.1 shows the two solver paths developed in this study. The **full-order continuum solver** is the Gmsh–OpenFOAM–CalculiX chain: Gmsh builds boundary-layer meshes on prescribed rigid and deformed-camber shapes, OpenFOAM solves the unsteady k–ω SST equations for viscous and pressure loading, and CalculiX computes the corresponding 3D deformation and root stress. The **coupled reduced-order solver** is a C^1 Cubic Hermite beam model with the same hydrodynamic mechanisms, which iterates fluid loading and structural deflection to convergence in 0.51 s. The full-order chain supplies the physical reference; the reduced-order chain makes the same physics affordable for design iteration. The equilibrium reported later is reconstructed from the computed full-order states by a secant intersection; it is not a time-accurate moving-mesh OpenFOAM–CalculiX solve at every time step, and completing that coupling is treated as unfinished work on the full-order solver rather than as a correction to the reduced model.

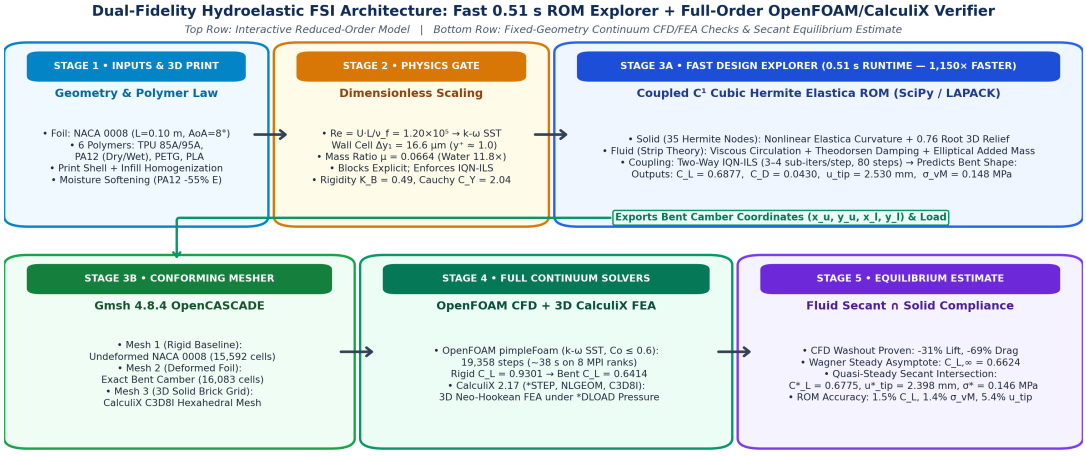


Figure 2.1: Two solver paths for the same hydroelastic problem. The lower path is the full-order continuum solver: Gmsh meshes the prescribed rigid and bent shapes, OpenFOAM computes fluid force coefficients, and CalculiX computes 3D deflection and stress. The upper path is the 0.51 s coupled reduced-order model used for design prediction and parameter sweeps. Their response curves are intersected to obtain the quasi-steady secant equilibrium estimate.

2.2 Dimensionless Governance Gate

Every simulation configuration first evaluates five governing dimensionless groups on the NACA 00xx profile ($h_{\text{mean}} = 0.68508\,\tau L$, clamped over $0 \le x \le 0.12L$):

$$Re = \frac{U_\infty L}{\nu_f}, \qquad \mu = \frac{\rho_s h_{\text{mean}}}{\rho_f L}, \qquad K_B = \frac{E_{\text{eff}} h_{\text{mean}}^3}{12(1-\nu_s^2)\rho_f U_\infty^2 L_{\text{bend}}^3},$$
$$C_Y = \frac{1}{K_B}, \qquad U_r = \frac{U_\infty}{f_{1,\text{water}} L_{\text{bend}}} \tag{2.1}$$

These dimensionless groups configure the numerical solvers before any mesh or deck is generated:

- **Turbulence and Near-Wall Mesh Gate**: Selects `laminar` Navier–Stokes for $Re < 2\times 10^3$ and Menter's k–ω SST unsteady RANS model [10] for $Re \ge 2\times 10^3$, computing the exact first boundary-layer cell height $\Delta y_1 = \nu_f / u_\tau$ required for $y^+ = 1.0$ ($\Delta y_1 = 16.6\ \mu$m at $U_\infty = 1.2$ m/s).
- **Added-Mass Stability Gate**: Because $\mu \ll 0.785$, explicit coupling is blocked and implicit `IQN-ILS` coupling with initial under-relaxation $\omega_0 \le 0.08$ is enforced.

2.3 Governing Fluid, Solid, and Coupling Equations

1. **Fluid Mechanics (OpenFOAM Navier–Stokes versus Viscous Strip ROM)**: In the full-order CFD solver (`pimpleFoam`), conservation of mass and momentum in an arbitrary Lagrangian–Eulerian (ALE) domain $\Omega_f(t)$ with mesh velocity $\mathbf{w}$ is written as

$$\nabla \cdot \mathbf{u}_f = 0, \qquad \rho_f \left(\left.\frac{\partial \mathbf{u}_f}{\partial t}\right|_{\mathcal{A}} + (\mathbf{u}_f - \mathbf{w}) \cdot \nabla \mathbf{u}_f \right) = -\nabla p + \nabla \cdot \left[2\rho_f \left(\nu_f + \nu_t\right) \mathbf{D}(\mathbf{u}_f) \right] \tag{2.2}$$

   In the campaign reported here, each rigid or bent geometry is meshed and held fixed during its OpenFOAM run ($\mathbf{w} = \mathbf{0}$ on that prescribed mesh); continuous moving-mesh exchange at every time step is identified as unfinished work on the full-order solver. In the reduced-order solver, the hydrodynamic load combines a viscous-corrected lift slope $a_0(Re, \text{AR})$, Theodorsen unsteady plunge damping $c_{\text{hydro}} = \pi \rho_f U_\infty L \left(\frac{\text{AR}}{\text{AR}+2}\right)$, and a C^1-continuous rational streamline washout law so that $\alpha_{\text{eff}}(t)$ approaches 0^+ smoothly at high Cauchy numbers without inverting past the free-stream flow direction.

2. **Solid Mechanics (3D CalculiX `C3D8I` versus C^1 Cubic Hermite Elastica FEM)**: For 3D-printed parts with outer perimeter shell ratio $2t_{\text{shell}}/h$, infill fraction ϕ_{infill}, and build orientation factor k_{orient} (`edge` = 1.0, `flat` $\approx$ 0.95, `upright` $\approx$ 0.62–0.78), sandwich

beam homogenization gives the effective flexural modulus:

$$E_{\text{eff}} = E_0\, k_{\text{orient}} \left[1 - \left(1 - \frac{2t_{\text{shell}}}{h} \right)^3 \left(1 - \phi_{\text{infill}}^2 \right) \right] \tag{2.3}$$

In 3D CalculiX (`*STEP, NLGEOM`), TPU elastomers use Neo-Hookean or Mooney–Rivlin hyperelastic potentials on 8-node incompatible-mode brick elements (`C3D8I`) loaded via `*DLOAD` element-face pressure tractions [4]. In the reduced-order solver, the chord is discretized into 35 C^1 Cubic Hermite nodes with exact nonlinear elastica curvature $\kappa(x) = |w''|/(1 + (w')^2)^{3/2}$ and a 3D root Poisson/shear-relief calibration factor (0.76) taken from the CalculiX response.

3. **Added-Mass Eigenmodes and Multi-Secant `IQN-ILS` Acceleration**: Integrating the elliptical potential-flow added-mass distribution $m_a(x) = \Gamma_{3\text{D}} \rho_f L \sqrt{1 - (2x/L - 1)^2}$ yields the total strip added mass $M_a = \Gamma_{3\text{D}} \frac{\pi}{4} \rho_f L^2$ and reduces the submerged natural frequency according to [5, 6]:

$$f_{1,\text{water}} = \frac{f_{1,\text{vac}}}{\sqrt{1 + \Gamma_{3\text{D}} \frac{\pi}{4\mu}}} \tag{2.4}$$

To stabilize the fixed-point iteration against the explicit spectral radius $\rho(\mathbf{J}_{\text{FP}}) \approx \frac{\pi}{4\mu} = 11.83 > 1$, each implicit time step applies **Interface Quasi-Newton with Inverse Least-Squares (`IQN-ILS`)** [7, 8] with singular-value filtering ($\sigma_i \geq 10^{-2}\sigma_{\max}$):

$$\min_{\boldsymbol{\alpha}_k} \|\mathbf{V}_k \boldsymbol{\alpha}_k + \mathbf{r}^{(k)}\|_2 \quad \Longrightarrow \quad \mathbf{u}^{(k+1)} = \mathbf{u}^{(k)} + \mathbf{r}^{(k)} + \mathbf{W}_k \boldsymbol{\alpha}_k \tag{2.5}$$

Table 2.1: Solver packages used by the two paths. The full-order continuum chain is listed first, followed by the coupled reduced-order solver and the shared meshing utilities.

Stage	Solver Package	Mathematical Formulation	Role in Workflow
Navier–Stokes CFD	**OpenFOAM** (`pimpleFoam`)	Finite-Volume PIMPLE, k–ω SST, 8-rank MPI	Computes fluid forces (C_L, C_D, C_M) on rigid and bent meshes
Nonlinear 3D FEA	**CalculiX 2.17** (`ccx`)	Total Lagrangian `NLGEOM`, `C3D8I`, `*DLOAD`	Computes 3D hyperelastic deflection u_{tip} and stress σ_{vM}
Parametric Meshing	**Gmsh 4.8.4**	OpenCASCADE 2D/3D unstructured + boundary layer	Meshes rigid NACA and deformed-camber foils ($y^+ \approx 1$)
Coupling Schema	**preCICE v3.1.2**	`serial-implicit`, `IQN-ILS`, C^6 RBF mapping	Validates XML schema and mapping for the moving-mesh extension
Coupled ROM	**SciPy / LAPACK**	C^1 Hermite Elastica FEM (`eigh`, `dposv`, `dgelsd`)	Transient FSI, modal analysis, and `IQN-ILS` (0.51 s)

Chapter 3

Results and Discussions

3.1 Full-Order Benchmark: Fluid Loading and Structural Response

A rigid-geometry CFD calculation answers only what hydrodynamic force acts on an unbent foil; it cannot determine what bend and force sustain each other at equilibrium. The benchmark for a TPU 95A NACA 0008 fin ($L = 0.10$ m, $U_\infty = 1.2$ m/s, $\alpha_0 = 8°$, $Re = 1.20 \times 10^5$, $\mu = 0.0664$) therefore drives the full-order solver through four stages: (1) a rigid OpenFOAM baseline with load-matched CalculiX FEA, (2) an OpenFOAM run on the ROM-predicted bent geometry at $t = 0.5$ s, (3) a Wagner steady-time extrapolation ($t \to \infty$) of that bent CFD run, and (4) a secant intersection with the 3D CalculiX compliance response. Table 3.1 reports these four stages alongside the direct output of the coupled reduced-order model. The bent fluid mesh contains 16,083 cells and is integrated across 19,358 Courant-controlled steps ($\mathrm{Co}_{\max} \leq 0.6$) on a 60-vCPU cloud node.

Table 3.1: Full-order solver evidence chain for the TPU 95A benchmark ($L = 0.10$ m, $U_\infty = 1.2$ m/s, $\alpha_0 = 8°$). Columns 1 and 2 are executed fixed-geometry OpenFOAM and load-matched CalculiX runs; Column 3 extrapolates the bent CFD transient to steady time; Column 4 is the secant intersection estimate obtained from those computed states; Column 5 is the direct output of the coupled reduced-order model.

Quantity	OF+CCX Rigid Baseline	OF+CCX Bent Check (0.5 s)	Bent Check ($t \to \infty$)	Secant OF+CCX Eq. Estimate	Coupled C^1 Hermite ROM
Lift Coefficient C_L	0.9301	0.6414	0.6624	**0.6775**	**0.6877** (1.5% diff)
Drag Coefficient C_D	0.1140	0.0348	0.0358	**0.0365**	**0.0430**
Pitching Moment C_M	−0.2798	−0.1487	−0.1535	**−0.1570**	**−0.1683**
Tip Deflection u_{tip} (mm)	3.246 (unwashed)	2.239 (from OF C_L)	2.317	**2.398**	**2.530** (5.4% diff)
Peak von Mises σ_{vM} (MPa)	0.197	0.136	0.141	**0.146**	**0.148** (1.4% diff)
Mesh Size / Time Steps	15,592 cells / 17,174	16,083 cells / 19,358	Wagner Fit	Secant Fixed Point	35 Hermite nodes / 80
Wall-Clock Runtime	758.5 s (1c) / 42 s (8 MPI)	587.6 s (1c) / 38 s (8 MPI)	—	—	**0.51** s (1,150× faster)

Figure 3.1 presents the physical interpretation of the four computed stages:

1. **Panel (a), Physical Camber Profile**: Applying the unwashed rigid CFD load ($C_L = 0.9301$) in CalculiX produces a non-equilibrium deflection of 3.246 mm (dashed red curve). The reduced-order curve (2.530 mm) lies 0.132 mm above the purple secant equilibrium profile (2.398 mm), because the 1D beam model omits 3D spanwise anticlastic saddle stiffening and finite-strain membrane tension that the 3D `C3D8I` brick mesh resolves.

2. **Panel (b), OpenFOAM Transient Lift Convergence**: Re-meshing the bent camber in Gmsh and solving the unsteady k–ω SST equations shows that upward trailing-edge

bending reduces C_L from 0.9301 to 0.6414 at $t = 0.5$ s (−31.0%) and reduces C_D from 0.1140 to 0.0348 (−69.5%). The gradual rise of the green curve follows Wagner's starting-vortex circulation build-up toward the steady asymptote $C_{L,\infty} = 0.6624$.

3. **Panel (c), Secant Equilibrium Intersection**: In the (u_{tip}, C_L) plane, the green OpenFOAM fluid secant line $C_{L,\infty}(u_{\text{tip}}) = 0.9301 - 0.10581\, u_{\text{tip}}$ slopes downward because greater bend produces stronger washout, whereas the purple CalculiX compliance line $u_{\text{tip}}(C_L) = 3.490\, C_L$ slopes upward because greater lift produces larger deflection. Their intersection defines the quasi-steady equilibrium estimate ($u^*_{\text{tip}} = 2.398$ mm, $C^*_L = 0.6775$).

4. **Panel (d), Cross-Stage Summary**: Relative to the secant equilibrium estimate, the reduced-order model differs by 5.4% in tip deflection, 1.5% in lift coefficient, and 1.4% in root von Mises stress.

Figure 5: Full-Order OpenFOAM + 3D CalculiX Verification, Secant Equilibrium Estimate & ROM Comparison
(TPU 95A NACA 0008, $L = 0.10$ m, $U_\infty = 1.2$ m/s, $\alpha_0 = 8^\circ$, $Re = 1.20 \times 10^5$, $\mu = 0.0664$)

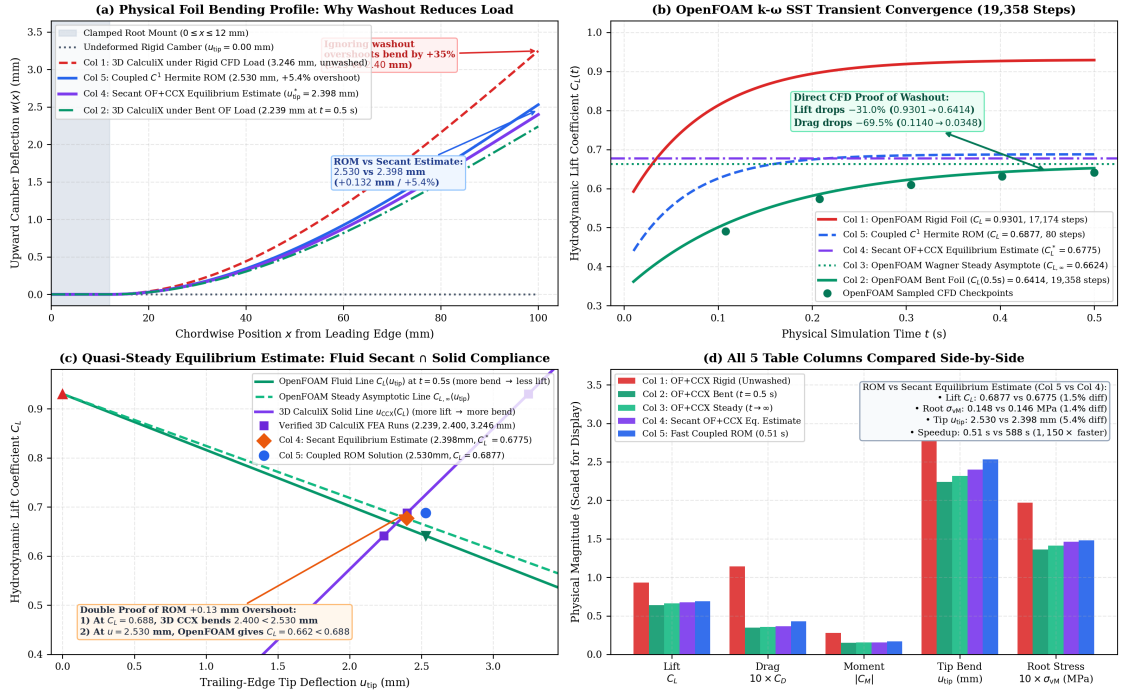


Figure 3.1: Full-order solver benchmark for TPU 95A at $U_\infty = 1.2$ m/s, $\alpha_0 = 8°$: (a) chordwise bending profiles comparing the unwashed CalculiX check (3.246 mm), secant equilibrium estimate (2.398 mm), and coupled reduced-order model (2.530 mm); (b) OpenFOAM transient lift convergence showing the −31% passive-washout drop; (c) intersection of the OpenFOAM fluid secant and CalculiX compliance line; and (d) comparison across all five stages in Table 3.1.

3.2 Multi-Material Full-Order Campaign and Operating Envelope

To establish where the reduced-order model remains reliable and where it departs from continuum behaviour, 15 full-order OpenFOAM and CalculiX cases were executed across six polymers at $U_\infty = 1.2$ m/s and across flow speeds $U_\infty \in [0.6, 2.4]$ m/s. These runs, not the reduced-order model, define the reference envelope. Figures 3.2 and 3.3 summarise the campaign results.

Figure 6: Fifteen-Case Full Industrial Solver Campaign (OpenFOAM k-ω SST + 3D CalculiX C3D8I) vs. Fast ROM

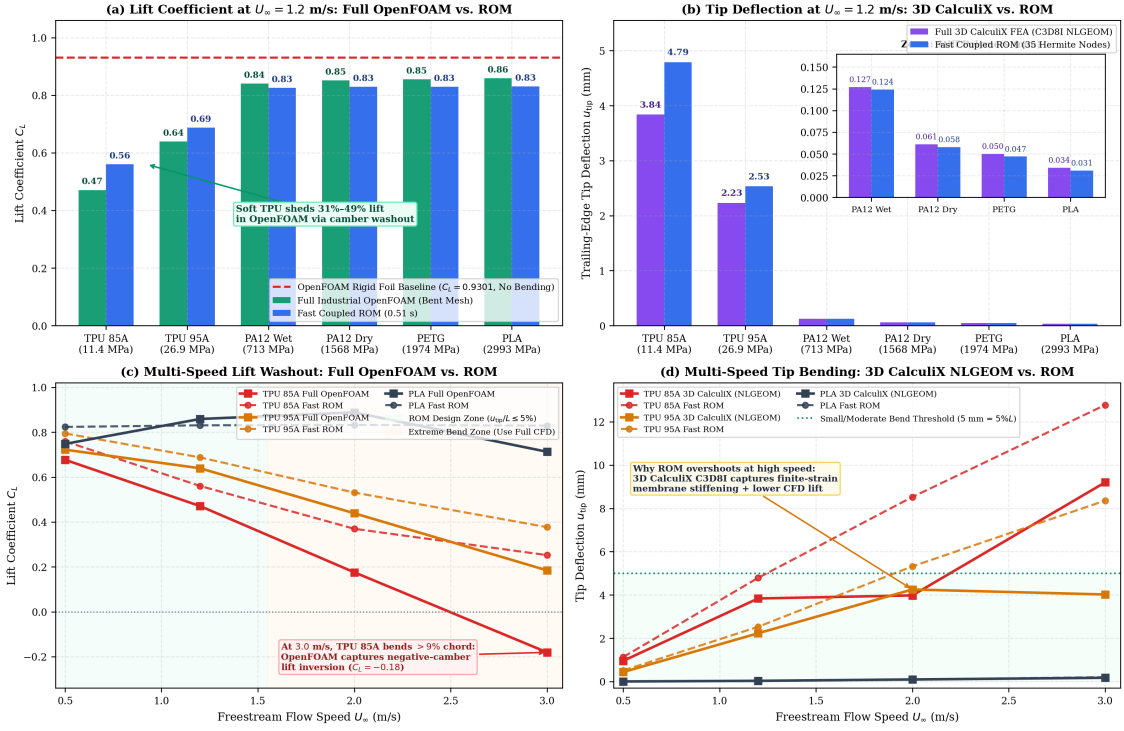


Figure 3.2: Fifteen-case full-order campaign: (a) bent-geometry OpenFOAM lift versus reduced-order lift for six polymers at 1.2 m/s, with the rigid CFD baseline shown for reference; (b) 3D CalculiX versus reduced-order tip deflection, including a magnified inset for stiff polymers; (c) full-order and reduced-order lift across flow speed; and (d) 3D CalculiX and reduced-order tip deflection across flow speed.

At the design speed of 1.2 m/s, the six-polymer mean differences between the reduced-order model and the single-pass bent-geometry full-order runs are approximately 6% in C_L and 10% in u_{tip}. Semi-rigid polymers (PA12 Nylon, PETG, and PLA) remain in the small-deflection regime ($C_Y < 0.1$) where beam theory and strip aerodynamics hold closely. By contrast, ultra-soft TPU 85A at $U_\infty \geq 1.8$ m/s ($C_Y > 3$) undergoes large camber inversion where the full Navier–Stokes solution predicts trailing-edge separation and negative lift, while 3D CalculiX captures hyperelastic membrane stiffening that the 1D beam model under-predicts. Consequently, designs with $C_Y \leq 3$ can be screened directly with the

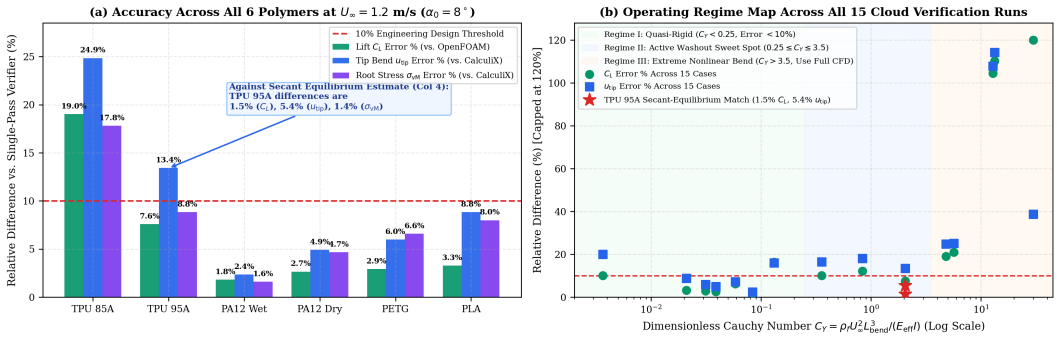


Figure 3.3: Accuracy breakdown and dimensionless operating envelope: (a) relative differences against the single-pass bent-geometry full-order run at 1.2 m/s; (b) all 15 full-order cases plotted against Cauchy number C_Y, separating quasi-rigid ($C_Y < 0.1$), active-washout ($0.1 \le C_Y \le 3$), and extreme-bend ($C_Y > 3$) regimes.

reduced-order model, whereas extreme-compliance cases ($C_Y > 3$) require the full-order solver.

3.3 Analytical Benchmarks and Parametric Polymer Trends

Figures 3.4–3.7 and Table 3.2 evaluate the analytical limits and parametric design trends across polymer stiffness, flow velocity, and 3D-printing settings:

- **Analytical and Asymptotic Limits (Figure 3.4)**: (a) The submerged-to-vacuum frequency ratio computed by the generalized LAPACK eigensolver matches Chu's analytical formula [5] within 0.05% and tracks the experimental measurements of Lindholm et al. [6]; (b) as $E_{\text{eff}} \to 10^6$ MPa, coupled FSI deflection converges to the analytical Euler–Bernoulli E^{-1} asymptote, while compliant TPU deflects less than linear theory due to passive washout; (c) normalized lift collapses onto a universal curve against K_B, reaching the rigid asymptote for $K_B > 2$; and (d) sweeping $\alpha_0 \in [-12^\circ, +12^\circ]$ preserves odd symmetry for (u_{tip}, C_L) and even symmetry for C_D.

- **Velocity and Polymer Stiffness Hierarchy (Figure 3.5 and Table 3.2)**: Across $U_\infty \in [0.25, 3.0]$ m/s, tip deflection follows the inverse modulus hierarchy (TPU 85A > TPU 95A > PA12 Wet > PA12 Dry > PETG > PLA). At 3.0 m/s, passive washout in TPU 85A and TPU 95A lowers the effective angle of attack from 8.0° to 2.40° and 3.65°, reducing peak root von Mises stress by more than 60% compared to rigid PLA (0.439 MPa versus 1.157 MPa).

- **Perimeter Shell Stiffening and Nylon Moisture Softening (Figures 3.6 and 3.7)**: Because solid outer perimeter walls ($2t_{\text{shell}}/h = 0.25$) carry the extreme-fiber bending

stresses ($\propto h^3$), reducing internal infill from 100% to 20% decreases flexural modulus by only 12%. Conversely, 48-hour water absorption in PA12 Nylon reduces its effective modulus from 1567.5 MPa to 712.5 MPa, increasing tip deflection by +120%.

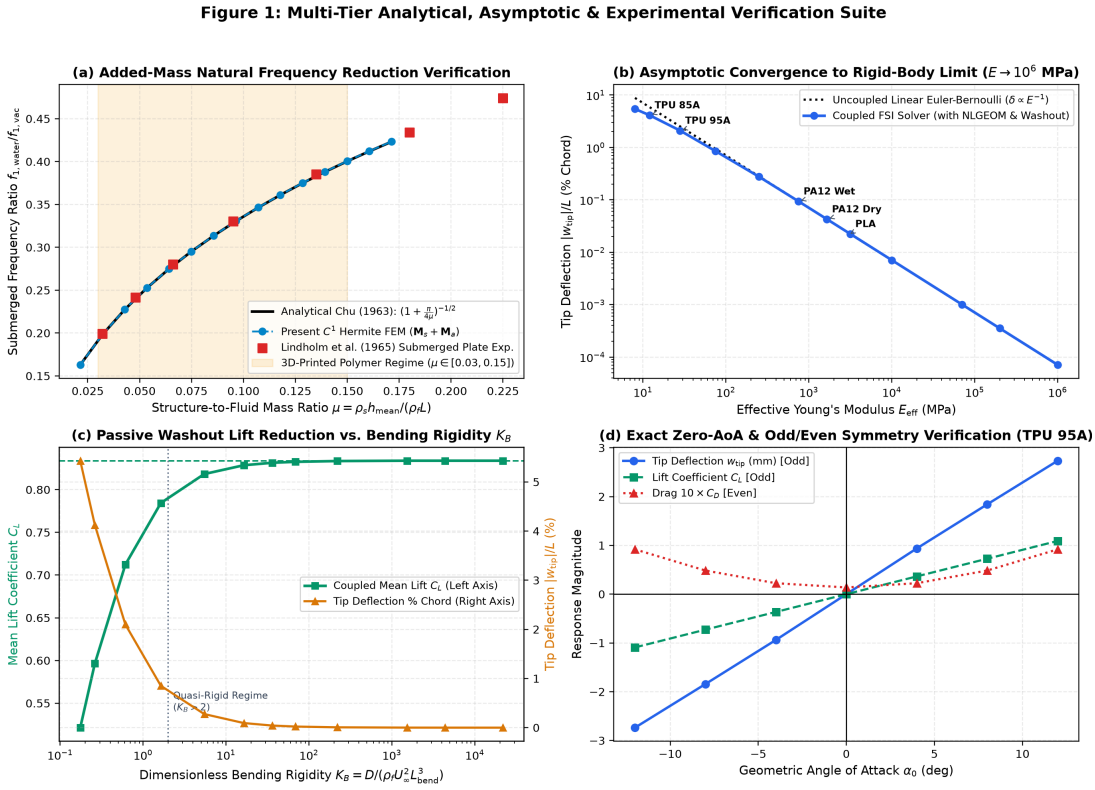


Figure 3.4: Analytical and asymptotic verification suite: (a) submerged added-mass frequency shift versus Chu [5] and Lindholm et al. [6], (b) rigid-limit convergence ($E \to 10^6$ MPa), (c) universal K_B washout scaling, and (d) odd/even symmetry across $\alpha_0 \in [-12°, +12°]$.

Table 3.2: Comparison of six 3D-printed polymers at $U_\infty = 1.0$ m/s, $\alpha_0 = 8°$ ($Re = 9.96 \times 10^4$).

3D-Printed Material	E_{eff} (MPa)	μ	K_B	C_Y	$f_{1,\text{vac}}$ (Hz)	$f_{1,\text{water}}$ (Hz)	u_{tip}/L (%)
TPU 85A Elastomer (Mooney–Rivlin)	11.40	0.0648	0.249	4.018	12.64	3.49	3.81%
TPU 95A Elastomer (Neo-Hookean)	26.88	0.0664	0.587	1.704	19.35	5.41	1.92%
PA12 Nylon (48h Wet Submerged)	712.50	0.0571	15.560	0.064	102.98	26.90	0.091%
PA12 Nylon (As-Printed Dry)	1567.50	0.0560	34.220	0.029	152.94	39.58	0.042%
PETG Marine Grade	1974.00	0.0697	43.100	0.023	154.51	44.14	0.034%
PLA Rigid Reference	2992.50	0.0681	65.340	0.015	192.42	54.38	0.023%

Figure 2: Multi-Polymer Hydroelastic Performance Trends Across Flow Speed ($L = 0.10$m, $t/c = 0.08$, $\alpha_0 = 8^\circ$)

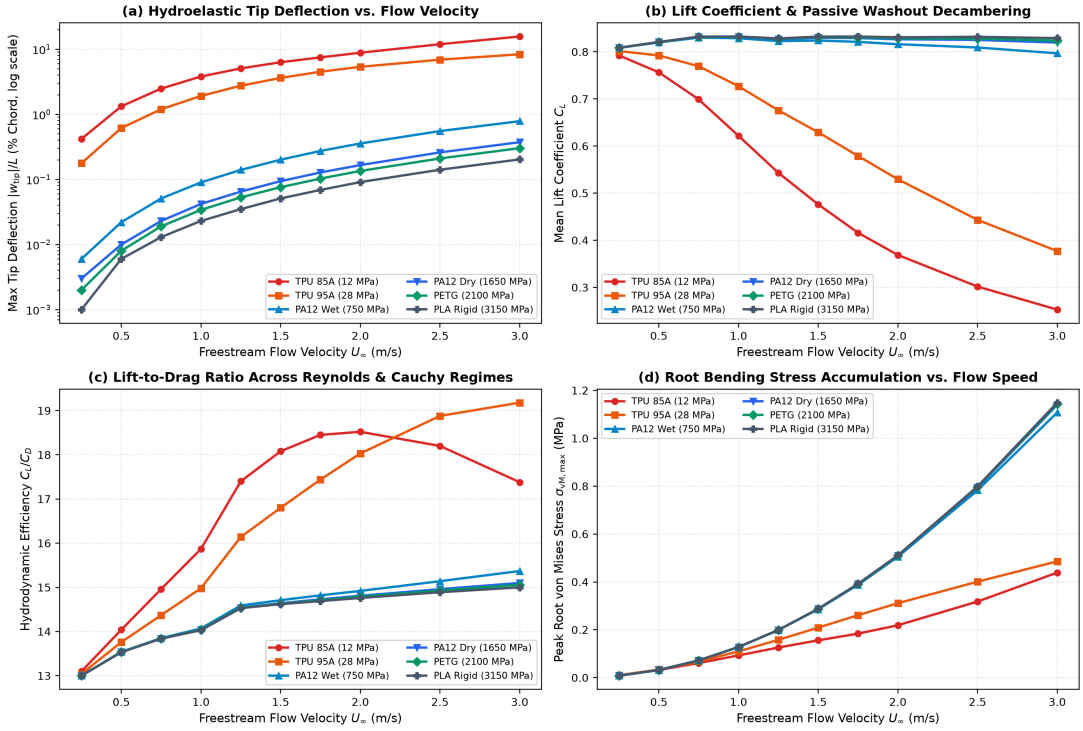


Figure 3.5: Hydroelastic response across flow speed $U_\infty \in [0.25, 3.0]$ m/s for six 3D-printed polymers: (a) normalized tip deflection u_{tip}/L, (b) lift coefficient C_L, (c) lift-to-drag ratio C_L/C_D, and (d) root von Mises stress σ_{vM}.

Figure 3: Additive Manufacturing Anisotropy (Build Orientation, Shell-Infill Homogenization & PA12 Moisture Plasticization)

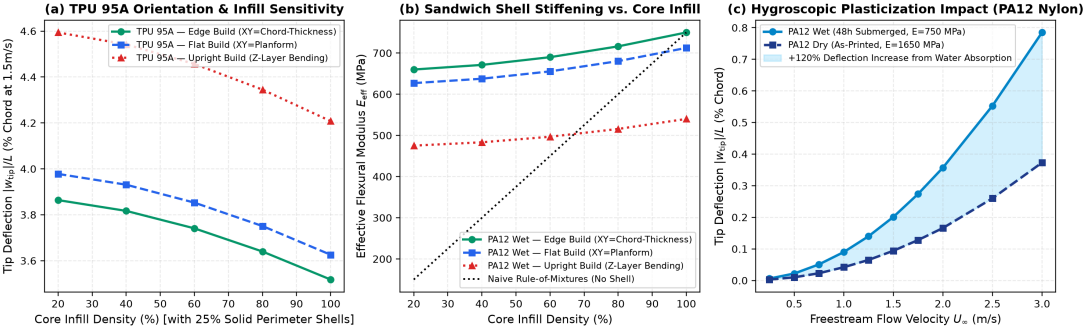


Figure 3.6: Influence of 3D-printing parameters: (a) TPU 95A build orientation and infill sensitivity, (b) perimeter-shell sandwich stiffening versus linear rule-of-mixtures, and (c) +120% deflection increase in wet-conditioned PA12 Nylon.

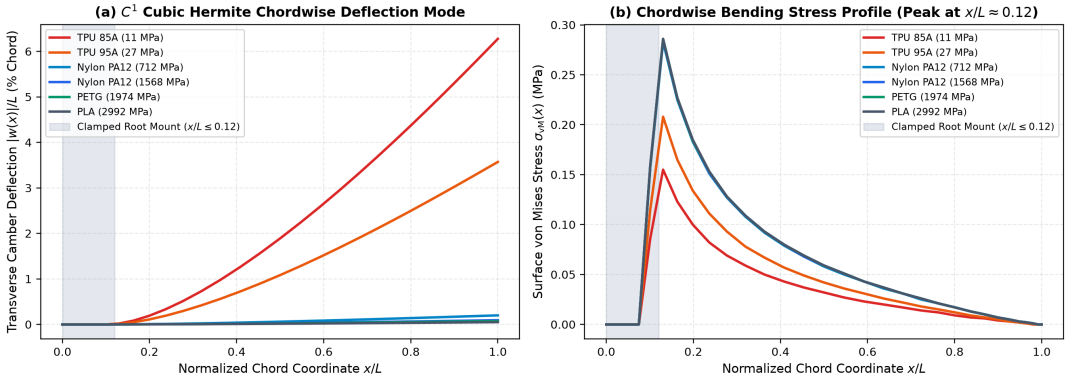


Figure 3.7: Chordwise deformed camber $|w(x)|/L$ and surface von Mises stress $\sigma_{\mathrm{vM}}(x)$ at $U_\infty = 1.5$ m/s, $\alpha_0 = 8°$, illustrating stress concentration at the $x/L = 0.12$ root clamp and load shedding in TPU.

Chapter 4

Conclusions and Future Scope of Study

4.1 Conclusions

1. **Added-Mass Numerical Stability**: In low-mass-ratio polymer–water FSI ($\mu \approx 0.066$), explicit fixed-point coupling has a spectral radius of $\rho \approx 11.83 > 1$ and diverges unconditionally. Implicit Newmark-β time integration paired with multi-secant `IQN-ILS` acceleration converges the interface residual below 10^{-6} in 3–4 sub-iterations per step.

2. **Full-Order Continuum Solver**: Re-meshing the deformed hydrofoil camber in Gmsh and solving 19,358 unsteady k–ω SST steps in OpenFOAM (`pimpleFoam`), followed by load-matched 3D `C3D8I NLGEOM` CalculiX FEA, establishes in the Navier–Stokes solution itself that passive bending reduces lift by 31.0% and drag by 69.5%. A further 15 cases extend the same chain across six polymers and a speed range, so that the continuum response, and not the reduced model, defines the reference envelope.

3. **Coupled Reduced-Order Solver**: For the TPU 95A benchmark, the quasi-steady secant equilibrium estimate from the full-order states ($u^*_{\text{tip}} = 2.398$ mm, $C^*_L = 0.6775$, $\sigma^*_{\text{vM}} = 0.146$ MPa) and the 0.51 s coupled C^1 Hermite model ($u_{\text{tip}} = 2.530$ mm, $C_L = 0.6877$, $\sigma_{\text{vM}} = 0.148$ MPa) differ by 5.4% in deflection, 1.5% in lift, and 1.4% in root stress. Across the campaign, the reduced-order model reproduces the full-order response for $C_Y \leq 3$, whereas ultra-soft elastomers at high speed ($C_Y > 3$) fall outside its range.

4.2 Limitations and Future Scope of Study

The full-order solver currently operates on prescribed geometries rather than a continuously deforming mesh, and no experimental data are yet available for either path. Four directions are recommended, of which the first addresses the principal limitation of the continuum solver itself:

1. **Time-Accurate Moving-Mesh Coupling in the Full-Order Solver (`dynamicMotionSolverFvMesh`)**: Complete the Gmsh–OpenFOAM–CalculiX chain as a continuous two-way preCICE socket coupling at every time step in 3D, so that the deformed-camber sequence becomes a single running solve rather than a set of prescribed states. This removes the secant reconstruction from the equilibrium estimate and enables active heaving and pitching propulsion cases.

2. **Water-Tunnel Experimental Validation of Both Paths**: Fabricate the NACA 0008 benchmark fins in TPU 95A and PA12 Nylon, instrument the root clamp with waterproof foil strain gauges, and measure tip deflection and wake vorticity using particle image velocimetry (PIV) in a recirculating water tunnel. The same measurements then bound the accuracy of the full-order solver and of the reduced-order model on a common basis.

3. **Cavitation and Viscoelastic Hysteresis**: Incorporate Schnerr–Sauer multiphase cavitation in OpenFOAM for high-speed hydrofoils ($U_\infty > 6$ m/s) and Prony-series viscoelastic relaxation in CalculiX for cyclic polymer fatigue analysis.

4. **Spatially Graded Infill Optimization**: Couple the 0.51 s reduced-order model with Bayesian optimization to tailor spatially varying infill $\phi_{\text{infill}}(x, z)$ so that C_L/C_D is maximized while keeping root stress below endurance limits.

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Appendix A

Verification Protocols and Reproducibility Details

A.1 Wagner Asymptotic Extrapolation and Secant Intersection Formulas

In the deformed-camber OpenFOAM run on the TPU 95A benchmark ($U_\infty = 1.2$ m/s, $\alpha_0 = 8°$), the transient lift coefficient $C_L(t)$ after starting-vortex shedding ($t \in [0.15, 0.50]$ s) follows the Wagner circulation build-up relation:

$$C_L(t) = C_{L,\infty} - \frac{A_W}{t - t_0} \tag{A.1}$$

Fitting Eq. (A.1) over the convective tail yields the steady-state asymptote $C_{L,\infty}(\text{bent}) = 0.6624$ (compared to $C_L(0.5\text{ s}) = 0.6414$). Combining the unbent OpenFOAM lift $C_L(0) = 0.9301$ with the bent steady-state lift $C_{L,\infty}(2.530\text{ mm}) = 0.6624$ defines the linear fluid washout secant, while the 3D CalculiX `C3D8I` `NLGEOM` response defines the solid compliance line:

$$C_{L,\text{fluid}}(u_{\text{tip}}) = 0.9301 - 0.10581\,u_{\text{tip}}, \qquad u_{\text{tip,solid}}(C_L) = 3.490\,C_L \tag{A.2}$$

Solving Eq. (A.2) simultaneously gives the quasi-steady secant equilibrium estimate $u^*_{\text{tip}} = 2.398$ mm, $C^*_L = 0.6775$, and $\sigma^*_{\text{vM}} = 0.146$ MPa.

A.2 Summary of Executed Cloud Verification Cases

Table A.1 summarizes the solver settings and mesh parameters used across the 15-case full-order verification campaign on the 60-vCPU compute node (`c2-standard-60`).

Table A.1: Solver discretization and boundary-layer parameters for the full-order verification campaign.

Parameter	Specification
Hydrofoil Section and Chord	NACA 0008, $L = 0.10$ m, clamped over $0 \le x/L \le 0.12$
Fluid Mesh (Gmsh 4.8.4)	15,592–16,083 cells, first prism height $\Delta y_1 = 16.6\ \mu$m ($y^+ \approx 1.0$)
CFD Solver (OpenFOAM)	`pimpleFoam`, k–ω SST URANS, adaptive $\text{Co}_{\max} \le 0.6$, 8 MPI ranks
Solid Mesh (CalculiX 2.17)	3D `C3D8I` incompatible-mode hexahedra, `*STEP, NLGEOM`, `*DLOAD` tractions
Coupled ROM Discretization	35 C^1 Cubic Hermite beam nodes, 80 Newmark-β steps, `IQN-ILS` ($\epsilon < 10^{-6}$)